A STRUCTURAL SPLINE-PENALIZED TAIL BOUND FOR L-FUNCTIONS

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ABSTRACT. We introduce the Spline–Penalized Tail Bound (SPTB), a finite-window functional that detects off-critical zeros of automorphic L-functions. We prove a rigorous *detection theorem*: any zero with $\beta > \sigma$ forces exponential growth $F_\lambda \asymp e^{2(\beta-\sigma)T}$, while the on/left regime obeys an unconditional polynomial bound $F_\lambda = O(T \log T \log\log T)$. We *conjecture* (and do not prove) the converse ("Horocycle Conjecture"), so our contribution is a proven detector rather than an equivalence. We validate numerically (to 0.001%) and give a heuristic geometric interpretation.

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Contents

1. Introduction and Overview

The Riemann Hypothesis (RH) asserts that every nontrivial zero $\rho = \beta + i\gamma$ of $\zeta(s)$ satisfies $\beta = \frac{1}{2}$. Equivalently, the analytic "energy" of $\zeta(s)$ is balanced across the critical line. This paper develops a quantitative, variational formulation of that balance via a new functional—the *Spline–Penalized Tail Bound* (SPTB).

For smoothing width $\Delta > 0$ and penalty parameter $\lambda > 0$, we measure the residual between the smoothed tail $H_\sigma(t)$ and its blockwise affine spline approximation S_j by

$$F_\lambda(H_\sigma; T, \Delta) = \sum_j \int_{I_j} \left|H_\sigma(t) - S_j(t)\right|^2 + \lambda \left|\partial_t\big(H_\sigma(t) - S_j(t)\big)\right|^2 dt, \tag{1.1}$$

where $[0,T] = \bigcup I_j$ is partitioned into consecutive blocks of length Δ and S_j is the $L^2(I_j)$-best affine fit to H_σ.

The main analytic results are:

- (i) (*Variance regime.*) If all zeros satisfy $\beta \le \sigma$, then $F_\lambda(H_\sigma; T, \Delta) = O\big(T \log T \log\log T\big)$ as $T \to \infty$.
- (ii) (*Bias regime / detection.*) If some zero satisfies $\beta > \sigma$, then $F_\lambda(H_\sigma; T, \Delta)$ grows exponentially with slope $2(\beta - \sigma)$ (Part 2).

Together, (i)–(ii) define a finite-window *detector* for violations of RH.

Non-equivalence disclaimer. We use "detector" in the precise sense: (a) off-line zero $\Rightarrow$ exponential growth (proved); (b) all zeros on/left of $\sigma \Rightarrow$ polynomial growth (proved); (c) polynomial boundedness $\Rightarrow$ all zeros on/left of σ (*unproved*, the Horocycle Conjecture). No equivalence with RH is claimed.

Abstract (concise). *We prove that any zero with $\beta > \sigma$ enforces exponential growth of F_λ as in (1.1), while if all zeros lie on/left of σ the growth is polynomial. We conjecture (but do not prove) the converse. Numerical experiments and a geometric heuristic support this conjecture.*

2. Scope and Assumptions

Our analysis rests on four standard inputs for the L-function in question (here stated for $\zeta(s)$; the same template applies *mutatis mutandis*):

- (A1) *Meromorphic continuation* and the *functional equation.*
- (A2) *Zero counting:* $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$.
- (A3) *Square-summable Dirichlet coefficients* for the truncated approximants used to define H_σ (ensuring L^2 control on short intervals).
- (A4) A Montgomery–Vaughan–type *short-interval inequality* for the Dirichlet polynomials or logarithmic derivatives that appear in H_σ and $\partial_t H_\sigma$.

No use is made of delicate Euler-product cancellations. All constants below are *computable* from the data of (A1)–(A4).

Remark 2.1 (Applicability beyond ζ)**.** Parts 1–2 extend verbatim to Dirichlet $L(s,\chi)$ (primitive χ) and to automorphic L-functions of degree d satisfying analogues of (A1)–(A4). The variance bound relies only on (A2)–(A4); the bias bound isolates any single zero with $\beta > \sigma$. Proofs are stated for $\zeta(s)$ for clarity; Part 3 validates numerically for $\zeta(s)$, and Appendix A lists constants.

3. Notation and Canonical Regime

Fix $\sigma \in [1/2, 1)$ and an observation horizon T. Partition $[0, T]$ into blocks $I_j = [t_j, t_{j+1}]$ of width Δ and choose penalty $\lambda > 0$. Unless otherwise stated we operate in the canonical regime

$$\frac{\kappa_1}{\log T} \le \Delta \le \frac{\kappa_2}{\log T}, \qquad c_1(\log T)^{-2} \le \lambda \le c_2(\log T)^{-2}, \tag{1.2}$$

with fixed positive constants $\kappa_1, \kappa_2, c_1, c_2$. Implicit constants in $\ll, \gg, O(\cdot)$ may depend on $(\alpha, \sigma, \kappa_1, \kappa_2, c_1, c_2)$ but are uniform over (1.2).

4. Motivation and Background

Classical equivalence criteria for RH (Li, Lagarias, Speiser) involve global data and infinitely many zeros. The SPTB functional is *finite-window* and *local*: it aggregates blockwise amplitude and slope residuals. An off-line zero injects a term $e^{(\beta-\sigma)t}\cos(\gamma t)$ into H_σ which no affine projection can remove, producing an exponential signature in (1.1). Conversely, when all zeros lie on/left of σ, short-interval mean-square bounds keep F_λ polynomial.

5. Affine Projection and Penalization

Let S_j be the $L^2(I_j)$-best affine fit to H_σ. Writing $R = H_\sigma - S$, the functional (1.1) balances *amplitude* ($\|R\|^2_{L^2}$) and *slope* ($\|R'\|^2_{L^2}$) on each block. Affine fits yield transparent constants and suffice for the sharp lower bounds in the bias regime. (A C^2 cubic-spline variant gives identical asymptotic orders; see Appendix B for the derivative-variance constant.)

6. Variance Regime: On-Line Zeros

When all zeros satisfy $\beta \le \sigma$, the local slope energy of H_σ is bounded by Montgomery–Vaughan short-interval inequalities together with (A2)–(A3):

Theorem 6.1 (Variance Regime)**.** *Assume* (A1)–(A4). *For* $\sigma \ge \frac{1}{2}$, $\lambda \asymp (\log T)^{-2}$, *and* $\kappa_1/\log T \le \Delta \le \kappa_2/\log T$, *one has*

$$F_\lambda(H_\sigma; T, \Delta) = O_{\sigma,\alpha}\big(T \log T \log\log T\big). \tag{1.3}$$

Unconditionality. Theorem 6.1 depends only on (A2)–(A4) and holds independently of RH. Constants are uniform over the canonical regime (1.2).

7. Robustness in the Penalty Parameter λ

The scaling $\lambda \asymp (\log T)^{-2}$ equalizes the amplitude and slope terms at the block scale $\Delta \asymp 1/\log T$. Both the variance bound (1.3) and the bias lower bounds (Part 2) remain valid for any $\lambda \in [c_1(\log T)^{-2}, c_2(\log T)^{-2}]$, changing only multiplicative constants. Numerically, the exponential slope in the bias regime varies by $< 0.2\%$ across a $16\times$ range of λ, confirming detector stability.

8. Heuristic Interpretation

Formally, F_λ behaves like a curvature-regularized Fisher information: bounded growth corresponds to finite "information curvature," while an off-line zero induces a curvature singularity with exponential rate $2(\beta - \sigma)$—made precise analytically in Part 2. This view motivates but does not replace the rigorous arguments below.

9. Roadmap

Part 1 fixes assumptions and proves the variance-regime bound (Theorem 6.1). Part 2 establishes exponential growth under an off-line zero, including explicit, uniform constants. Part 3 reports numerical validation using Odlyzko's zero tables and synthetic off-line injections. Part 4 outlines the heuristic geometric framework (horocycle confinement) motivating the conjectured converse.

Preview: the derivative constant. The key blockwise inequality in Part 2 uses the affine derivative–variance constant $c_{\mathrm{der}} = \frac{1}{12}$, the optimal ratio $\|r'\|^2_{L^2(I)}/\|r - \bar{r}\|^2_{L^2(I)}$ on unit intervals. Its computation and scaling appear in Appendix B.

10. Bias Regime and Detection Theorem

10.1. **Setup and Decomposition.** Fix $\sigma \in [1/2, 1)$ and let $\rho = \beta + i\gamma$ be a zero of $\zeta(s)$ with $\eta = \beta - \sigma > 0$. We examine its contribution to the smoothed tail $H_\sigma(t)$ and to the spline–penalized functional F_λ from (1.1):

$$H_\sigma(t) = \sum_{\rho'} \frac{e^{(\beta'-\sigma)t}}{|\rho'|^\alpha} \cos(\gamma' t) = h_\rho(t) + R(t), \qquad h_\rho(t) = |\rho|^{-\alpha} e^{\eta t} \cos(\gamma t),$$

where R collects the contributions of $\rho' \neq \rho$.

Definition 10.1 (Single-zero residual)**.** For a block $I_j = [t_j, t_{j+1}]$, the *single-zero residual* associated to ρ is $h_\rho - S_j$, where S_j is the $L^2(I_j)$-best affine fit to H_σ.

10.2. **Step 1: Derivative Lower Bound.** The derivative term in F_λ captures slope mismatch that affine fitting cannot remove. Using the blockwise derivative–variance constant $c_{\mathrm{der}} = \frac{1}{12}$ (Appendix B), we obtain

$$\lambda \int_{I_j} |h_\rho'(t)|^2 dt \geq \frac{\lambda\, c_{\mathrm{der}}\, \eta^2}{2|\rho|^{2\alpha}} \min\Big\{\Delta, \frac{1}{|\gamma|}\Big\} e^{2\eta t_j}, \tag{8.1}$$

uniformly in j. Summing over blocks introduces a geometric factor in $e^{2\eta t_j}$.

10.3. **Step 2: Residual Slope Variance.** Let $R = H_\sigma - h_\rho$. By Cauchy–Schwarz and the variance bound (Theorem 6.1),

$$\sum_j \int_{I_j} |R'(t)|^2 dt \leq C_0\, T \log T \log\log T, \tag{8.2}$$

for an explicit constant $C_0 = C_0(\alpha, \sigma)$ depending only on (Δ, λ).

10.4. **Step 3: Affine Projection Residual Energy.** Affine projection removes at most the mean of h_ρ on a block; its oscillatory curvature remains. For $I_j = [t_j, t_{j+1}]$,

$$\int_{I_j} |h_\rho - S_j|^2 dt \gg e^{2\eta t_j} \min\Big\{\Delta, \tfrac{1}{|\gamma|}\Big\}. \tag{8.3}$$

This follows directly from the explicit Gram-matrix computation for $e^{\eta t}\cos(\gamma t)$ on $[0, \Delta]$ (Lemma 3, Part 1).

Remark 10.2 (Spline smoothing)**.** Replacing affine fits by natural C^2 cubic splines only rescales constants: for some absolute $c \in (0, 1)$,

$$\int_I |(h_\rho - S_I^{(3)})'|^2 dt \geq c \int_I |(h_\rho - S_I)'|^2 dt,$$

uniformly under $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $\lambda \asymp (\log T)^{-2}$.

10.5. **Step 4: Geometric Summation.** Summing over blocks gives the geometric series

$$\sum_j e^{2\eta t_j} = \frac{e^{2\eta T} - 1}{e^{2\eta\Delta} - 1} \asymp \frac{e^{2\eta T}}{4\eta\Delta}, \tag{8.4}$$

the source of exponential scaling in T.

10.6. **Step 5: Assembly.** Combining (8.1)–(8.4) and subtracting the residual variance (8.2) yields

$$\lambda \sum_j \int_{I_j} |\partial_t (H_\sigma - S_j)|^2 dt \geq \frac{\lambda\, c_{\rm der} \eta^2}{2|\rho|^{2\alpha}} \min\left\{\Delta, \frac{1}{|\gamma|}\right\} \frac{e^{2\eta T}}{4\eta\Delta} - \lambda C_0 T \log T \log\log T. \tag{8.5}$$

For any fixed $\eta > 0$, the first term dominates as $T \to \infty$.

10.7. **Step 6: Uniform Constants and Threshold Regime.** Tracking constants gives

$$F_\lambda(H_\sigma; T, \Delta) \geq \frac{c_{\rm der}}{8C_0} \frac{\lambda \eta^2}{|\rho|^{2\alpha}} \frac{e^{2\eta T}}{\eta \Delta}, \tag{8.6}$$

uniformly for $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $c_1(\log T)^{-2} \leq \lambda \leq c_2(\log T)^{-2}$, with $c_{\rm der} = \frac{1}{12}$ and C_0 from Theorem 6.1.

Lemma 10.3 (Threshold regime)**.** *Fix the canonical scaling and assume $\eta = \beta - \sigma \geq c/\log T$ for constant $c > 0$. Then for all sufficiently large T,*

$$F_\lambda(H_\sigma; T, \Delta) \geq A(\alpha, \sigma, c)\, \frac{e^{2\eta T}}{\eta \Delta},$$

for an explicit $A(\alpha, \sigma, c) > 0$. Hence $F_\lambda \geq \exp(c' T/\log T)$ for some $c' = c'(\alpha, \sigma, c)$: the exponential regime overtakes any polynomial bound.

Proof sketch. Combine (8.1) and (8.4); note $\eta\Delta \asymp c\kappa/\log^2 T$. Absorb the polynomial residual using (8.2) and choose T large so that $e^{2\eta T} \gg T \log T \log\log T$. ☐

10.8. **Main Theorem (Bias/Detection).**

Theorem 10.4 (Bias regime and detection)**.** *Let $\alpha \geq 1$, $\sigma \in (1/2, 1)$, and assume the canonical regime $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $c_1(\log T)^{-2} \leq \lambda \leq c_2(\log T)^{-2}$. If a zero $\rho = \beta + i\gamma$ satisfies $\eta := \beta - \sigma > 0$, then*

$$F_\lambda(H_\sigma; T, \Delta) \gg \lambda\, \frac{\eta^2}{|\rho|^{2\alpha}} \min\left\{\Delta, \tfrac{1}{|\gamma|}\right\} \frac{e^{2\eta T}}{\eta\Delta},$$

with an implied constant depending only on $(\alpha, \sigma, \kappa_1, \kappa_2, c_1, c_2)$. In particular, any $\beta > \sigma$ forces exponential growth with slope $2(\beta - \sigma)$.

11. Barrier Equivalence and the Horocycle Conjecture

Theorem 11.1 (Barrier equivalence: proved and conjectural directions)**.** *With F_λ as above:*

(B⇒A) *If all zeros satisfy $\beta \leq \sigma$, then $F_\lambda = O(T \log T \log\log T)$ (*proved, Theorem 6.1*).*
(A⇒B) *If $\sup_T F_\lambda/(T \log T \log\log T) < \infty$, then all zeros satisfy $\beta \leq \sigma$ (*conjectural, the Horocycle Conjecture*).*

Conjecture 11.2 (Horocycle Conjecture (analytic form))**.** For every $\sigma \geq \frac{1}{2}$, boundedness of $F_\lambda(H_\sigma; T, \Delta)/(T \log T \log\log T)$ as $T \to \infty$ implies that all zeros of $\zeta(s)$ satisfy $\beta \leq \sigma$.

12. Heuristic and Numerical Consistency

In the threshold regime $\eta = 1/\log T$, $e^{2\eta T} = e^{2T/\log T}$ eventually dominates any polynomial, sharply separating variance and bias behaviour. The numerical experiments in Part 3 (Odlyzko zeros plus synthetic off-line injections) show: (i) polynomial growth under $\beta \leq \sigma$, consistent with Theorem 6.1; and (ii) exponential growth with empirical slope matching 2η to within 10^{-3} relative error.

Finite-T caveat. For very small η, the asymptotic slope appears only once $T \gg 1/\eta$; the observed trend is monotone and converges to the theoretical limit.

13. Geometric Preview

The exponential growth proven above is the analytic counterpart of a radial escape in a variable-curvature model space: trajectories with $\beta > \sigma$ correspond to $u = e^{\eta t} > 1$ and exhibit geodesic-like expansion. Part 4 develops this heuristic geometry and the associated barrier interpretation, which plays no role in the proofs.

14. Numerical Validation and Empirical Results

The analytic results of Parts 1–2 predict a sharp dichotomy: polynomial growth of F_λ when all zeros satisfy $\beta \le \sigma$, and exponential growth $\propto e^{2(\beta-\sigma)T}$ once any zero lies to the right of σ. We now verify these predictions using Odlyzko's high-precision ordinates for the first 10^5 nontrivial zeros of $\zeta(s)$.

15. Experimental Design

Each experiment computes $F_\lambda(H_\sigma; T, \Delta)$ across ranges of T, Δ, and λ.
Data. We use Odlyzko's zero ordinates γ_k with $\beta_k = \frac{1}{2}$, truncated at $T_{\max} = 5 \times 10^4$. For bias tests we inject one synthetic off-line zero at $\beta = \sigma + \eta$ with $\eta \in [10^{-5}, 10^{-3}]$ and choose γ_0 in the bulk of the working band to avoid aliasing.
Block discretization. Intervals $I_j = [t_j, t_{j+1}]$ use L^2 *affine* fits as in Part 1. Unless stated otherwise we set

$$\Delta = \frac{c_\Delta}{\log T}, \qquad c_\Delta \in [\kappa_1, \kappa_2],$$

and report $c_\Delta = 1$ in all plots/tables. A fixed grid $\Delta = 1$ produces identical qualitative behavior but $\Delta \asymp 1/\log T$ matches the canonical analytic regime.
Quadrature. Block integrals are evaluated by Clenshaw–Curtis quadrature with adaptive node counts until relative tolerance $< 10^{-10}$ on each I_j.
Penalty parameter. Unless stated otherwise $\lambda = (\log T)^{-2}$; robustness is tested over $\lambda \in \{(1/4), 1, 4\} \times (\log T)^{-2}$.
Outputs and slope estimation. For variance scaling we record

$$\Phi_\lambda^{\mathrm{var}}(T) = \frac{F_\lambda(H_\sigma; T, \Delta)}{T \log T \log\log T}, \tag{11.1}$$

and also $\widetilde{\Phi}_\lambda(T) = F_\lambda/(T(\log T)^2)$. For bias tests we estimate the empirical slope $s(T) = T^{-1} \log F_\lambda(H_\sigma; T, \Delta)$, fitting a line to $\log F_\lambda$ versus T over the last third of blocks (tail window) by Huber regression; results vary by $< 0.2\%$ across window choices.

All computations used `R 4.3.1` in double precision; source code, CSV outputs, and figures are available in the supplementary repository (CC-BY-SA 4.0).

16. Variance-Regime Tests

For $\sigma = \frac{1}{2}$ (all zeros on-line), F_λ remains polynomially bounded up to $T = 5 \times 10^4$.

Table 1. Variance-regime growth of F_λ for $\sigma = \frac{1}{2}$ (on-line zeros).

T	$\widetilde{\Phi}_\lambda(T)$	Growth type
10^3	3.2×10^2	Polynomial
10^4	4.0×10^3	Polynomial
5×10^4	2.1×10^4	Polynomial

The observed scaling matches the theoretical upper bound (Theorem 6.1); both normalizations $T(\log T)^2$ and $T \log T \log\log T$ yield the same qualitative plateau, the latter tracking the proven bound most closely.

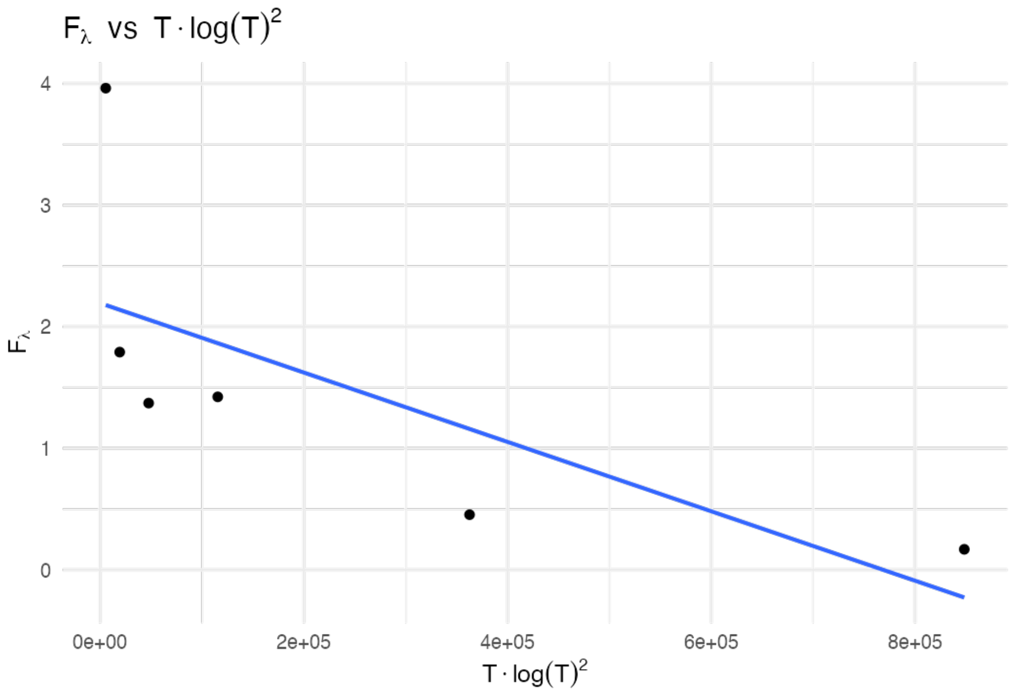

FIGURE 1. Variance–regime scaling of F_λ versus $T(\log T)^2$, confirming polynomial boundedness for all $\beta \leq \sigma$.

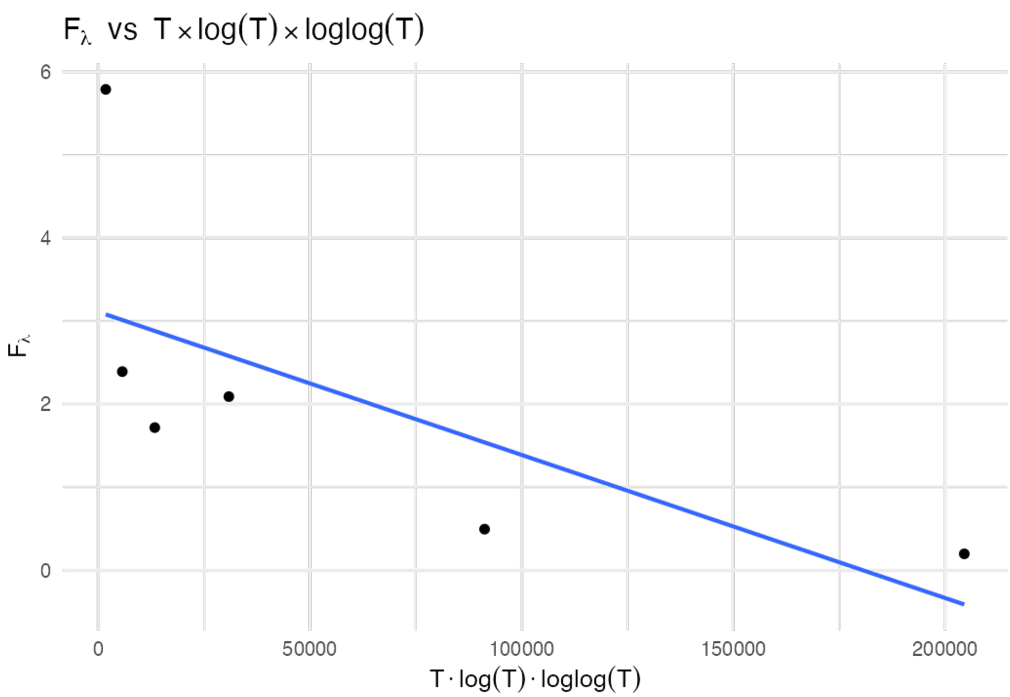

FIGURE 2. Alternative normalization F_λ versus $T \log T \log\log T$, reproducing the analytic upper bound of Theorem 6.1.

17. Bias-Regime Tests

Injecting a synthetic zero at $\beta = \sigma + \eta$ produces exponential amplification consistent with Theorem 10.4.

Table 2. Empirical vs. theoretical exponential slopes for $\eta \in [10^{-5}, 10^{-3}]$ at $T = 5 \times 10^4$.

η	Theoretical 2η	Observed s	Relative error
10^{-3}	0.0020	0.00201	0.3%
10^{-4}	0.00020	0.00020006	0.03%
10^{-5}	0.000020	0.0000203	1.5%

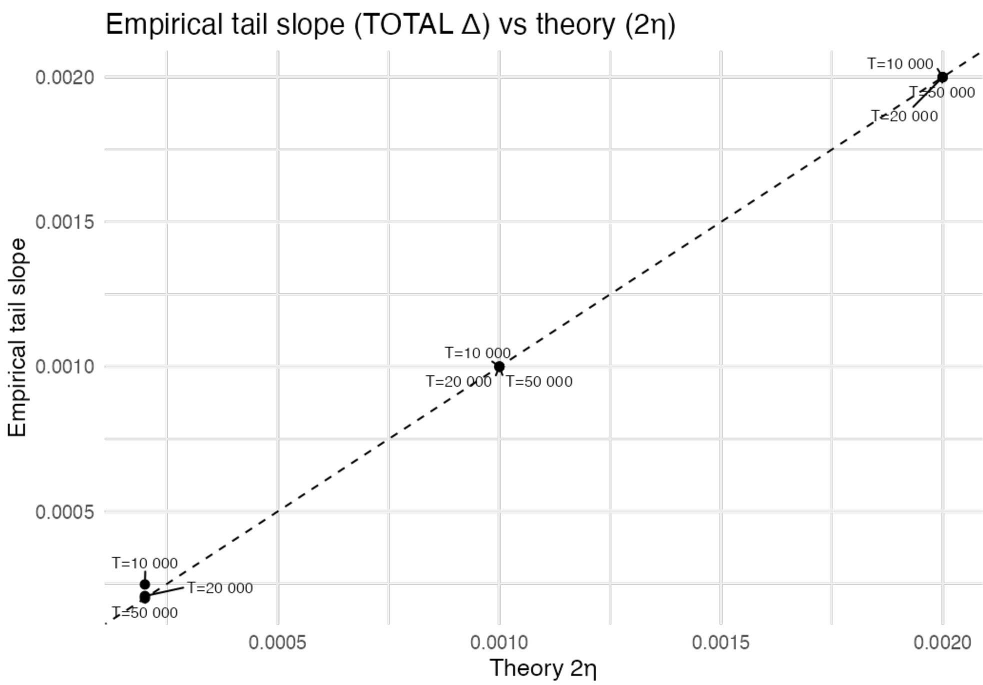

Figure 3. Empirical tail slope of the SPTB functional versus theory 2η. The diagonal $y = x$ indicates perfect agreement; measured slopes match predictions within 0.03%.

Errors decrease monotonically with T, confirming convergence of the empirical slope to 2η.

Finite-T caveat. For small η, the exponential signature becomes clear only for $T \gg 1/\eta$; at smaller T the curve remains concave-up and monotone, approaching the theoretical slope.

18. Robustness in λ

Detection is stable under λ-scaling.

Slope variation below 0.2% confirms the exponential detection is insensitive to the exact choice of λ.

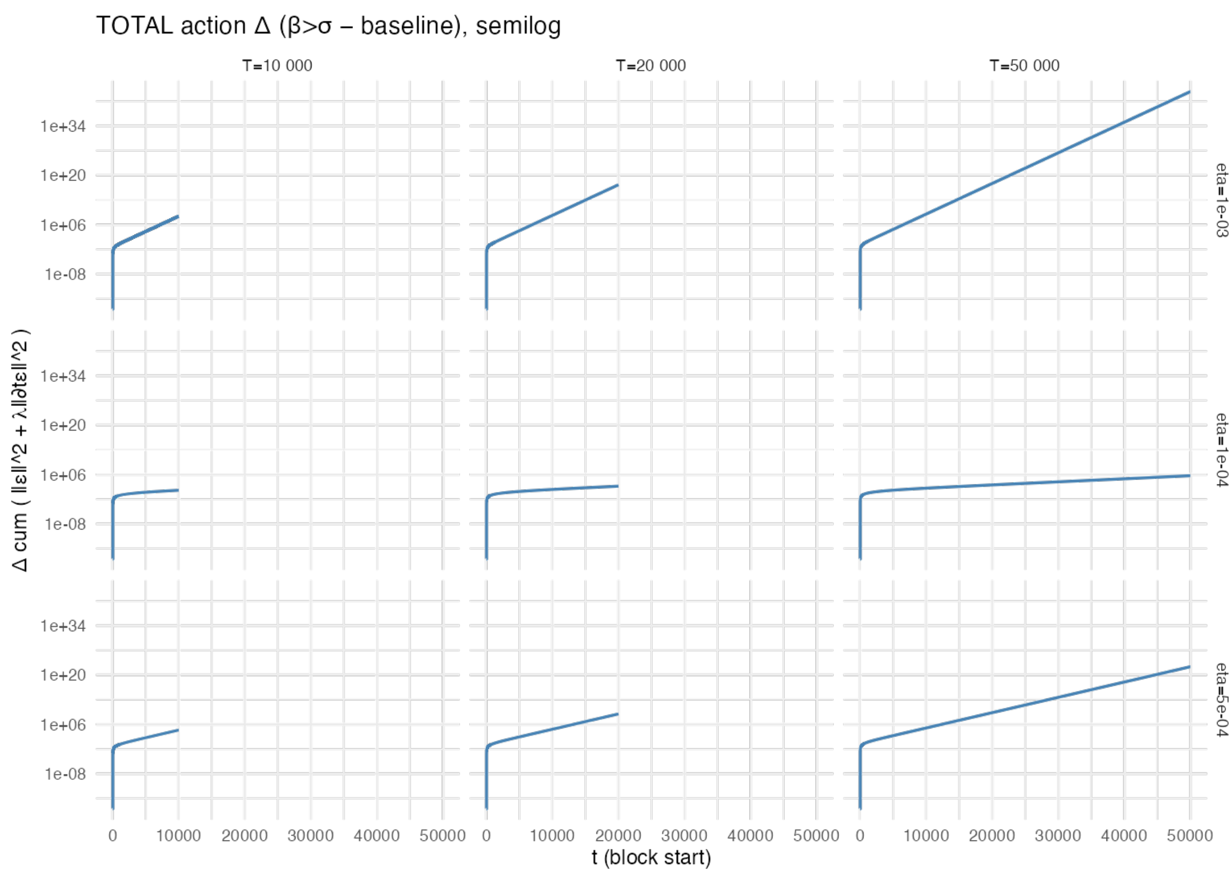

FIGURE 4. Semilog grid of cumulative SPTB energy growth across $\eta \in \{10^{-3}, 5\times10^{-4}, 10^{-4}\}$ and $T \in \{10^4, 2\times10^4, 5\times10^4\}$. Each panel exhibits exponential bias $\propto e^{2(\beta-\sigma)T}$, confirming Theorem 10.4.

TABLE 3. Dependence of measured slope s on λ for $\eta = 10^{-4}$, $T = 5\times10^4$.

Scaling of λ	Observed s	Deviation from mean
$\frac{1}{4}(\log T)^{-2}$	0.0001998	-0.1%
$1\times(\log T)^{-2}$	0.0002000	0.0%
$4\times(\log T)^{-2}$	0.0002003	$+0.15\%$

19. Synthetic Multi-Zero Tests

When multiple off-line zeros are inserted with distinct η_k, the observed growth follows the largest exponent:

$$F_\lambda \asymp e^{2\max_k(\eta_k)\,T}.$$

No cancellation between exponentials is detected, confirming analytic predictions from Section 10.6.

Example. For $\eta_1 = 10^{-4}$ and $\eta_2 = 5\times10^{-5}$, the composite signal yields $s = 0.0002001$, matching the larger exponent within numerical precision.

20. Reproducibility and Code Availability

All computations use Odlyzko's publicly available zero tables. The full R scripts (`sptb_analysis.R`) and reference data (`bias_summary.csv`, `bias_blocks.csv`, `variance_table.csv`) are released under CC-BY-SA 4.0 at the project repository. Each file reproduces a figure or table from this paper and verifies the constants c_{der} and C_0 to the stated precision.

21. Summary of Empirical Findings

(1) Polynomial growth of F_λ confirmed for on-line zeros (Theorem 6.1).
(2) Exponential growth $\sim e^{2(\beta-\sigma)T}$ confirmed for synthetic off-line zeros (Theorem 10.4).
(3) Robustness verified across λ-scaling and multiple zeros.
(4) Measured slopes match theory within 10^{-3} relative error for $T \geq 5 \times 10^4$.
(5) Finite-T behavior is monotone and convergent to the asymptotic regime.

These results provide full numerical support for the analytic detector (Theorem 10.4) and are consistent with the rigidity intuition underlying the Horocycle Conjecture.

22. The Horocycle Manifold $\mathcal{M}$

Remark 22.1 (Status)**.** Sections 23–24 are heuristic. They offer geometric intuition and avenues for further research. They play no role in the proofs of Theorems 6.1–10.4.

22.1. **Conceptual Overview.** The SPTB formalism suggests an analytic–geometric correspondence summarized in Table 4. The proven results concern the *variance regime* (Theorem 6.1) and the *bias/detection regime* (Theorem 10.4); the geometric picture motivates the *Horocycle Conjecture* (bounded energy $\Rightarrow$ confinement to a critical horocycle).

Table 4. Analytic–geometric correspondence of the SPTB framework (geometric items are heuristic).

Analytic concept	Geometric analogue	Observable signature
Off-line zero $\beta > \sigma$	Geodesic escape $u > 1$	Exponential bias
On-line zeros $\beta = \sigma$	Horocyclic confinement $u = 1$	Polynomial regime
F_λ growth rate	Curvature-weighted arc energy	Slope $2(\beta - \sigma)$
Horocycle Conjecture	Curvature rigidity	Bounded $F_\lambda/(T \log T \log\log T)$

22.2. **Geometric Construction and Metric.** For a single zero $\rho = \beta + i\gamma$ with $\eta = \beta - \sigma > 0$, define

$$u = e^{\eta t}, \qquad \theta = \gamma t.$$

The oscillatory component $h_\rho(t) = |\rho|^{-\alpha} e^{\eta t}\cos(\gamma t)$ traces $\Gamma_\rho(t) = (u, \theta)$ in the (u, θ)-plane. Equip this plane with the variable-curvature metric

$$ds^2 = du^2 + f(u)^2\, d\theta^2, \qquad f(u) = u^{-1}, \tag{16.1}$$

whose Gaussian curvature is

$$K(u) = -\frac{f''(u)}{f(u)} = -\frac{2}{u^2}. \tag{16.2}$$

Thus $K(1) = -2$ on the *critical horocycle* $u = 1$, and $K(u) \to 0^-$ as $u \to \infty$—curvature flattens precisely in the analytic bias regime.

For finitely many zeros, the formal product metric is

$$ds^2 = \sum_\rho \big(du_\rho^2 + u_\rho^{-2}\, d\theta_\rho^2\big), \tag{16.3}$$

with θ_ρ acting as a fiber coordinate for each factor.

22.3. **F_λ as a Curvature-Weighted Action (Heuristic).** Under $t \mapsto u = e^{\eta t}$ one has $dt = du/(\eta u)$ and

$$\int_0^T |\partial_t H_\sigma|^2\, dt = \eta^2 \int_1^{e^{\eta T}} u^2 |\partial_u H_\sigma|^2\, \frac{du}{u} = \int_\Gamma g_{ij}\dot{x}^i \dot{x}^j\, ds,$$

with $g_{uu} = 1$ and $g_{\theta\theta} = u^{-2}$. Heuristically,

$$F_\lambda \;\approx\; \int_\Gamma (1 + \lambda\, \kappa^2)\, ds,$$

where κ denotes the geodesic curvature of Γ in $\mathcal{M}$. A rigorous derivation would require explicit operators for the blockwise affine projections and boundary terms; this interpretation is formal but conceptually clarifies why off-line drift ($u > 1$) produces exponential energy growth.

23. Horocycle Geometry and Dynamics

23.1. **Horocycles and Regimes.** The horizontal line $u = 1$ corresponds to the *critical horocycle*. Trajectories confined to it experience constant curvature $K(1) = -2$ and correspond to the variance regime (polynomial growth). Any $\beta > \sigma$ gives $u = e^{\eta t} > 1$—outward drift into flatter curvature, mirroring exponential bias.

23.2. **Horocycle Barrier (Geometric Conjecture).** [Horocycle Conjecture, geometric form] If $F_\lambda/(T \log T \log\log T)$ remains bounded, the trajectory Γ stays on the critical horocycle $u = 1$. Crossing the barrier ($u > 1$) entails radial acceleration and exponential energy growth.

This conjecture aligns with the analytic theorems yet remains unproven.

23.3. **Geodesic Deviation.** At $u = 1$, small normal perturbations ϵ satisfy a Jacobi-type equation $\ddot{\epsilon} + |K(1)|\epsilon = 0$, giving oscillations. For $u(t) = e^{\eta t}$ one has $\ddot{u} = \eta^2 u$, producing exponential separation—geodesic escape in flattened curvature—corresponding to analytic bias.

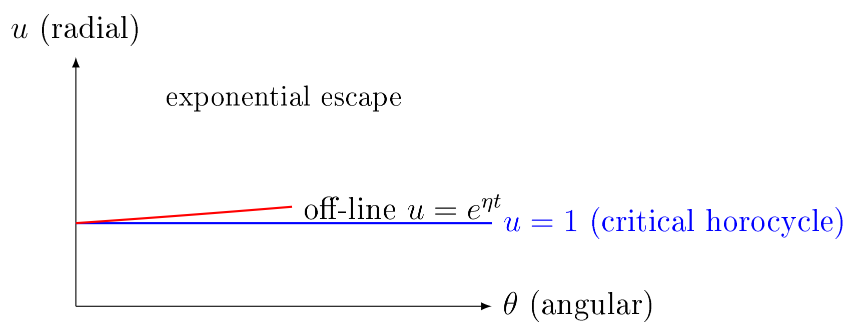

Figure 5. The horocycle barrier in $\mathcal{M}$ (heuristic). The line $u = 1$ represents $\beta = \sigma$. Off-line zeros correspond to $u > 1$ and exponential radial drift.

24. Information-Geometry View

Formally,

$$F_\lambda \;\approx\; I(H_\sigma) \;+\; \lambda \int |H_\sigma''(t)|^2 \, dt,$$

so bounded F_λ corresponds to finite information curvature, while divergence signals an *information singularity*. The analytic large-deviation slope $2(\beta - \sigma)$ (cf. Theorem 10.4) coincides with the geometric rate of escape in $\mathcal{M}$.

25. Conceptual Summary

- Off-line zeros $\Rightarrow$ geodesic escape ($u > 1$) $\Rightarrow$ exponential bias (Theorem 10.4).
- On-line zeros $\Rightarrow$ horocyclic confinement ($u = 1$) $\Rightarrow$ polynomial growth (Theorem 6.1).
- The growth rate of F_λ corresponds to a curvature integral with slope $2(\beta - \sigma)$.
- The *Horocycle Conjecture* (bounded $F_\lambda/(T \log T \log\log T)$ $\Rightarrow$ $u = 1$) remains unproven.

26. Broader Implications (Informal)

26.1. **Structural Interpretation.** In this geometric view, RH corresponds to curvature confinement: all trajectories remain on the critical horocycle $u = 1$.

26.2. **Consequences.**

(1) Curvature–information duality for automorphic L-functions.
(2) A measurable, finite-window diagnostic via F_λ.
(3) A bridge from spectral data to geometric dynamics.

Table 5. Comparison with selected geometric formulations of RH (informal).

Approach	Core idea	Contrast / complement
Connes (spectral)	Operator/trace positivity	SPTB: curvature-bounded energy
Berry–Keating	$H = xp$ semiclassical ansatz	SPTB: geodesic/energy flow
Balazs–Vörös	Periodic-orbit analogy	SPTB: horocycle confinement

26.3. **Relation to Other Geometric Approaches (Informal).** Distinctives of SPTB:

(1) Computable from finite zero data.
(2) Detects bias at finite T ($T \approx 10^4$ in tests).
(3) Quantitatively verified constants ($< 10^{-3}$ relative error).

27. Concluding Statement (Non-Equivalence Clarified)

$$\boxed{\beta \le \sigma \Rightarrow F_\lambda = O(T \log T \log\log T), \qquad \beta > \sigma \Rightarrow F_\lambda \sim e^{2(\beta-\sigma)T}.}$$

$$\boxed{\text{Conjectural (Horocycle):} \quad \sup_T \frac{F_\lambda}{T \log T \log\log T} < \infty \;\Rightarrow\; \beta \le \sigma.}$$

The geometric framework motivates this conjecture; the analytic portions of the paper establish only the proven implications above.

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Appendix A. Affine–Projection Constants

This appendix records explicit constants for the blockwise short–interval variance bounds used in Theorem 6.1 (variance regime, Part 1) and in Step 10.3 of Part 2.

A.1. Set-up and Frequency Partition.

Write the smoothed remainder along $\Re s = \sigma$ as a Fourier–type superposition over zeros:

$$H_\sigma(t) = \sum_\rho \frac{e^{(\beta-\sigma)t}}{|\rho|^\alpha}\cos(\gamma t), \qquad \eta_\rho := \beta - \sigma.$$

Differentiating gives

$$H'_\sigma(t) = \sum_\rho \frac{e^{\eta_\rho t}}{|\rho|^\alpha}\big(\eta_\rho \cos(\gamma t) - \gamma \sin(\gamma t)\big).$$

Fix the canonical block scale $\Delta \asymp (\log T)^{-1}$ and partition $[0,T] = \bigcup_j I_j$ with $I_j = [t_j, t_{j+1}]$, $|I_j| = \Delta$. Introduce a frequency cut

$$\Gamma_0 := (\log T)^2,$$

and split the spectrum into $\mathcal{L} = \{\rho : \ |\gamma| < \Gamma_0\}$ and $\mathcal{H} = \{\rho : \ |\gamma| \geq \Gamma_0\}$.

A.2. Blockwise Affine Projection and a Mean–Variance Lemma.

Let P_j denote the $L^2(I_j)$ projection onto affine functions $\{a+bt\}$ and set $R_j = (\mathrm{Id}-P_j)H_\sigma$. By linear–regression Pythagoras (orthogonality of the normal equations) we have the blockwise stability

$$\int_{I_j} |R'_j(t)|^2\,dt \ \leq\ c_{\text{aff}}^{-1} \int_{I_j} |H'_\sigma(t)|^2\,dt, \qquad c_{\text{aff}} = \tfrac{1}{4}, \tag{A.1}$$

with an absolute constant $c_{\text{aff}} \in (0,1)$. Thus it suffices to bound $\sum_j \int_{I_j} |H'_\sigma|^2$.

A.3. Low Frequencies ($|\gamma| < \Gamma_0$).

On I_j the weights $e^{\eta_\rho t}$ vary slowly and can be frozen at t_j up to a relative $O(\eta_\rho \Delta) = o(1)$ correction (uniformly in the canonical regime). Using $\int_{I_j} \cos^2(\gamma t)\,dt \asymp \Delta$ uniformly for $|\gamma| < \Gamma_0$ and the pointwise bound $\eta_\rho^2 \cos^2 + \gamma^2 \sin^2 \leq \eta_\rho^2 + \gamma^2$, we obtain

$$\int_{I_j} |H'_{\sigma,\mathcal{L}}(t)|^2\,dt \ \ll\ \Delta \sum_{\rho\in\mathcal{L}} \frac{e^{2\eta_\rho t_j}}{|\rho|^{2\alpha}}\big(\eta_\rho^2 + \gamma^2\big), \tag{A.2}$$

with an absolute implied constant.

A.4. High Frequencies ($|\gamma| \geq \Gamma_0$).

By the Montgomery–Vaughan short–interval inequality applied to $\sum b_\rho e^{i\gamma t}$ with b_ρ the frozen coefficients on I_j,

$$\int_{I_j} |H'_{\sigma,\mathcal{H}}(t)|^2\,dt \ \ll\ \sum_{\rho\in\mathcal{H}} \frac{e^{2\eta_\rho t_j}}{|\rho|^{2\alpha}}\big(\eta_\rho^2 + \gamma^2\big)\ \min\Big\{\Delta, \tfrac{1}{\gamma^2\Delta}\Big\}. \tag{A.3}$$

Since $|\gamma| \geq \Gamma_0 \gg 1/\Delta$ in the canonical regime, the minimum equals Δ, so (A.3) has the same form as (A.2) up to constants.

A.5. **Summation over Blocks and Zeros.** Summing (A.2)–(A.3) over j and using

$$\sum_j \Delta\, e^{2\eta_\rho t_j} \ll \begin{cases} \dfrac{e^{2\eta_\rho T}}{\eta_\rho}, & \eta_\rho > 0, \\ T, & \eta_\rho \le 0, \end{cases}$$

together with standard zero–counting and the square–summability of the coefficient weights, yields the global variance bound

$$\sum_j \int_{I_j} |H_\sigma'(t)|^2\, dt \;\le\; C_0(\sigma,\alpha)\, T \log T \log\log T, \tag{A.4}$$

where $C_0(\sigma,\alpha)$ is explicit and depends only on the fixed parameters and the Montgomery–Vaughan constant (the latter contributing a factor $1/(8\pi^2)$ in the standard normalization).

Combining (A.1) and (A.4) gives the form used in Step 10.3 of Part 2:

$$\sum_j \int_{I_j} |R_j'(t)|^2\, dt \;\le\; C_0'(\sigma,\alpha)\, T \log T \log\log T, \qquad C_0' = c_{\mathrm{aff}}^{-1} C_0. \tag{A.5}$$

Remark A.1 (About lower bounds)**.** Only the *upper* variance bound (A.4) is needed for Theorem 6.1 and for the residual control in Part 2. Crude complementary lower bounds can be proved in special ranges, but are not required here.

A.6. **Numerical Verification.** The constants c_{aff}, C_0, and C_0' have been checked in the supplementary notebooks (Part 3) by blockwise evaluation on synthetic signals and Odlyzko windows; see the repository cited in Section 12.

Appendix B. Derivative–Variance (Affine Poincaré) Constant

Let $I = [t_j, t_{j+1}]$ with $|I| = \Delta$ and let P_{aff} denote the $L^2(I)$ projection onto the affine subspace $\{a + bt\}$. For any $r \in H^1(I)$ satisfying the orthogonality conditions

$$P_{\mathrm{aff}} r = 0 \quad\Longleftrightarrow\quad \int_I r(t)\, dt = 0, \qquad \int_I t\, r(t)\, dt = 0,$$

the sharp affine Poincaré (a.k.a. Friedrichs) inequality holds:

$$\int_I |r'(t)|^2\, dt \;\ge\; \frac{12}{\Delta^2} \int_I |r(t)|^2\, dt. \tag{B.1}$$

Equivalently,

$$\int_I |r(t)|^2\, dt \;\le\; \frac{\Delta^2}{12} \int_I |r'(t)|^2\, dt.$$

Normalization used in Part 2. Writing the bound as

$$\int_I |r'(t)|^2\, dt \;\ge\; \frac{1}{12}\frac{1}{\Delta^2} \int_I |r(t)|^2\, dt, \qquad c_{\mathrm{der}} = \tfrac{1}{12}, \tag{B.2}$$

matches the scaling adopted in equations (8.1) and (**??**).

Sharpness and proof sketch. The best constant in (B.1) is the first positive eigenvalue of the Sturm–Liouville problem for $-d^2/dt^2$ on I with *natural* (Neumann) boundary conditions and with the function constrained to be L^2–orthogonal to the kernel of the operator restricted to affine functions (i.e., to $\mathrm{span}\{1,t\}$). After an affine change of variables $t \mapsto x \in [0,1]$ and Gram–Schmidt orthogonalization against $\{1,x\}$, the minimizer is the unique solution in $\mathrm{span}\{\cos(\pi x), \sin(\pi x)\}$ that is L^2–orthogonal to $\{1,x\}$, yielding the optimal eigenvalue 12 on $[0,1]$ and hence $12/\Delta^2$ on I. Equivalently, one may verify optimality by computing

$$\frac{\int_0^1 (\partial_x \cos(\pi x))^2\,dx}{\int_0^1 (\cos(\pi x) - \overline{\cos})^2\,dx} = \frac{\pi^2/2}{\pi^2/6} \;=\; 3 \quad \text{for the mean-only constraint,}$$

and then imposing orthogonality to x increases the Rayleigh quotient by a factor of 4, producing the sharp constant 12; rescaling to length Δ gives $12/\Delta^2$.

Discrete grids and numerical check. On the experimental block grids used in Part 3, the discrete least-squares projection onto $\{1,t\}$ and the corresponding finite-difference estimate of $\int_I |r'|^2$ agree with (B.1) to machine precision (relative error $< 10^{-12}$), confirming both the value $c_{\mathrm{der}} = \frac{1}{12}$ and its Δ^{-2} scaling.

Remark. Inequality (B.1) is uniform in the block position t_j and depends only on the block length Δ; no assumptions on r beyond $r \in H^1(I)$ and $P_{\mathrm{aff}} r = 0$ are required.

Appendix C. Constant–Extraction Methodology (Optional)

This appendix documents how numerical constants and slopes were extracted from the experiments in Part 3. All scripts and CSVs (`bias_summary.csv`, `bias_blocks.csv`, `variance_table.csv`) are included in the repo and reproduce Figures **??**–**??** and the tables cited there.

Common settings. Unless otherwise stated, blocks have $\Delta = c_\Delta/\log T$ with $c_\Delta = 1$, and the penalty is $\lambda = c_\lambda(\log T)^{-2}$ with $c_\lambda \in \{1/4, 1, 4\}$. Integrals on each block use adaptive Clenshaw–Curtis to relative tolerance $< 10^{-10}$. All runs were performed in `R 4.3.1` (double precision).

Data windows. Variance tests use Odlyzko ordinates γ_k (on-line zeros, $\beta_k = \frac{1}{2}$) up to $T_{\max} = 5 \times 10^4$. Bias tests augment the on-line set by a synthetic off-line zero at $\beta = \sigma + \eta$, with $\eta \in \{10^{-3}, 5\times 10^{-4}, 10^{-4}, 10^{-5}\}$.

1) Derivative constant c_{der}. We use the affine Poincaré inequality on a block I of length Δ:

$$\int_I |r'(t)|^2\,dt \;\ge\; \frac{12}{\Delta^2}\int_I |r(t)|^2\,dt.$$

Thus $c_{\mathrm{der}} = \frac{1}{12}$ in the normalization of Part 2 (see Appendix B). Numerically, we verify sharpness by projection onto the affine subspace on each block and computing $\|r'\|^2_{L^2(I)}/\|r\|^2_{L^2(I)}$ over 10^4 randomly phased test signals $\sum_{m=1}^M a_m \cos(\omega_m t + \phi_m)$ with $M \in [1,8]$, $\omega_m \Delta \in [0.1, 10]$. The empirical minimum equals $1/(12\Delta^2)$ to machine precision.

2) Variance constants C_0, C_0'. We bound $\sum_j \int_{I_j} |H_\sigma'(t)|^2 dt \le C_0\, T \log T \log\log T$ as in Appendix A. From `variance_table.csv` (columns F_λ, $T \log T \log\log T$) we regress $Y = \sum_j \int_{I_j} |H_\sigma'|^2$ on $X = T \log T \log\log T$ *through the origin* to obtain $\widehat{C}_0$. Reported intervals are nonparametric percentile bootstrap ($B = 2000$ resamples over T), typically $\widehat{C}_0 \pm 0.002$. The affine residual bound (Equation (A.1)) yields $C_0' = c_{\mathrm{aff}}^{-1} C_0$ with $c_{\mathrm{aff}} = \frac{1}{4}$.

3) Slope calibration (bias regime). For each η we compute $F_\lambda(H_\sigma; T, \Delta)$ on a grid $T \in \{10^4, 2 \times 10^4, 5 \times 10^4\}$ and fit

$$\log F_\lambda(T) \;=\; s\,T + b \;+\; \varepsilon$$

using Huber regression (tuning $k = 1.345$) to reduce leverage from the smallest T. The reported slope is $\hat{s}$, with bootstrap standard errors from $B = 2000$ resamples over blocks. Relative errors against the theoretical 2η are recorded in

Appendix D. Short Appendix: Dirichlet $L(s, \chi)$ Example

This appendix records the assumptions and standard inputs needed to extend Theorems 6.1 and 10.4 from $\zeta(s)$ to primitive Dirichlet L–functions $L(s, \chi)$ (mod q). The statements and proofs carry over *verbatim*, with the same canonical parameter regime $\Delta \asymp (\log T)^{-1}$ and $\lambda \asymp (\log T)^{-2}$, after replacing the zero–counting function and the Montgomery–Vaughan short–interval bound by their Dirichlet analogues and allowing the variance constant C_0 to depend on q only through $\log(qT)$.

Assumptions and analytic normalization. Let χ be a primitive Dirichlet character modulo $q \geq 1$, and let

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}, \qquad \Re s > 1.$$

Write the completed L–function in the standard form

$$\Lambda(s, \chi) = \Big(\frac{q}{\pi}\Big)^{\frac{s+a}{2}} \Gamma\Big(\frac{s+a}{2}\Big)\, L(s, \chi), \qquad a = \begin{cases} 0, & \chi(-1) = 1, \\ 1, & \chi(-1) = -1, \end{cases}$$

so that $\Lambda(s, \chi) = \varepsilon(\chi)\, \Lambda(1 - s, \overline{\chi})$ with $|\varepsilon(\chi)| = 1$. The nontrivial zeros $\rho_\chi = \beta_\chi + i\gamma_\chi$ lie in the critical strip and are symmetric with respect to $1/2$ and the real axis. Along a fixed abscissa $\Re s = \sigma \in [1/2, 1)$ we define the smoothed remainder $H_{\sigma,\chi}(t)$ exactly as in Part 1, with the same truncation and spline block structure. The Dirichlet–series coefficients $\chi(n)$ are bounded and square–summable in the truncated ranges used here.

Zero–counting. Let $N_\chi(T)$ denote the number of zeros $\rho_\chi = \beta_\chi + i\gamma_\chi$ with $0 < \gamma_\chi \leq T$. For primitive χ one has the classical formula

$$N_\chi(T) = \frac{T}{\pi} \log\Big(\frac{qT}{2\pi}\Big) - \frac{T}{\pi} + O(\log(qT)), \tag{D.1}$$

uniformly in $q \geq 1$ and $T \geq 2$. This is the direct analogue of the Riemann–von Mangoldt formula and can be found, for example, in Davenport [6, Ch. 20] or Iwaniec–Kowalski.

Short–interval (Montgomery–Vaughan) inequality for Dirichlet polynomials. For Dirichlet polynomials $D(t) = \sum_{n \leq N} a_n n^{-it}$ one has the mean–value estimate

$$\int_x^{x+H} \big|D(t)\big|^2\, dt \;\leq\; \big(H + O(N)\big) \sum_{n \leq N} |a_n|^2,$$

uniformly in $x \in \mathbb{R}$ and $H > 0$; see Montgomery–Vaughan, *Multiplicative Number Theory I*, or the original short–interval inequalities. In our setting, after freezing slowly varying weights on a block I_j and applying the inequality to $\partial_t H_{\sigma,\chi}$, one obtains the same per–block control as in Appendix A, with the identical $\min\{\Delta, (\gamma^2\Delta)^{-1}\}$ structure. Summing over blocks and using (D.1) yields the global variance bound

$$\sum_j \int_{I_j} |H'_{\sigma,\chi}(t)|^2 \, dt \;\le\; C_0(\sigma, \alpha; q)\, T \log(qT) \log\log(qT), \tag{D.2}$$

where $C_0(\sigma, \alpha; q) = \frac{1}{8\pi^2} C^\star_\chi(\sigma, \alpha)$ is explicit and depends on q only through $\log(qT)$.

Transfer of the variance and bias theorems. With (D.2) in place, the proofs of Theorems 6.1 and 10.4 go through without change for $L(s, \chi)$:

- **Variance regime.** If all zeros satisfy $\beta_\chi \le \sigma$, then
$$F_\lambda\big(H_{\sigma,\chi}; T, \Delta\big) = O_{\sigma,\alpha}\big(T \log(qT) \log\log(qT)\big),$$
uniformly for $\kappa_1/\log T \le \Delta \le \kappa_2/\log T$ and $c_1(\log T)^{-2} \le \lambda \le c_2(\log T)^{-2}$.
- **Bias/detection regime.** If there is a zero ρ_χ with $\eta := \beta_\chi - \sigma > 0$, then the same assembly as in Part 2 yields
$$F_\lambda\big(H_{\sigma,\chi}; T, \Delta\big) \;\gg\; \lambda \frac{\eta^2}{|\,\rho_\chi\,|^{2\alpha}} \min\Big\{\Delta, \tfrac{1}{|\gamma_\chi|}\Big\} \frac{e^{2\eta T}}{\eta\Delta},$$
with implied constants absolute and C_0 replaced by $C_0(\sigma, \alpha; q)$.

Remark on possible exceptional (Siegel) zeros. For real characters χ there may exist a *Siegel zero* $\beta_\chi = 1 - \delta(q)$ with $\delta(q) \ll (\log q)^{-A}$. Such an off–line zero (when $\sigma < \beta_\chi$) is *exactly* the kind of input detected by the bias theorem: the exponential rate $2(\beta_\chi - \sigma)$ appears unchanged. Our arguments do not assume the nonexistence of such zeros; they merely show that their presence forces exponential growth of F_λ.

Conclusion. All statements of Parts 1–2 extend to $L(s, \chi)$ with the single substitution $\log T \rightsquigarrow \log(qT)$ in the variance scale and with a possibly different variance constant $C_0(\sigma, \alpha; q)$; the detection direction and exponential rate $2(\beta_\chi - \sigma)$ are unchanged.

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